BRAINIAC

A Counterfactual-Guided Transformer Framework for Stability-Aware MRI Brain Tumor Risk Stratification

A

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CANDIDATE'S DECLARATION

We hereby certify that the project work entitled "Brainiac: A Counterfactual-Guided Transformer Framework for Stability-Aware MRI Brain Tumor Risk Stratification," submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering with specialization in AIML at the Faculty of Engineering and Technology, Jain (Deemed to be University), Bengaluru, is an authentic record of the work carried out by us under the guidance of Dr. Kanithan. The present report reflects the improved and technically refined version of the Brainiac project, incorporating the methodological improvisations discussed during the development process and the insights consolidated from the accompanying research paper.

The matter presented in this report has not been submitted, either in part or in full, for the award of any other degree, diploma, or academic certification at this or any other institution. All references to existing literature, datasets, models, and related frameworks have been duly acknowledged in the references section.

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ABSTRACT

Brain tumors remain a major challenge in neuro-oncology because accurate MRI-based analysis must support not only detection and segmentation but also reliable downstream decision-making. Although modern deep learning systems have substantially improved volumetric tumor segmentation, many existing approaches focus primarily on predictive accuracy and do not explicitly measure how stable a model's decision remains under small, anatomically plausible variations. This limitation is critical in safety-sensitive medical settings, where two visually similar cases may receive predictions with very different levels of robustness.

This report presents Brainiac, a stability-aware MRI brain tumor analysis framework that integrates Swin-UNETR-based three-dimensional segmentation, radiomic feature extraction, transformer attention entropy, tumor-constrained counterfactual generation, and downstream risk stratification. The central contribution of the framework is the Counterfactual Sensitivity Score (CSS), defined as the minimum perturbation required to alter the model's risk prediction while preserving anatomical plausibility within the tumor region of interest. CSS is not treated merely as a post-hoc explanation output; rather, it is incorporated as a quantitative feature alongside radiomic and attention-derived descriptors for low-, medium-, and high-risk grouping based on decision stability.

The improved Brainiac pipeline also provides dual explainability. Integrated Gradients offer spatial explanations showing which MRI regions contribute most strongly to predictions, while counterfactual MRI comparisons provide a causal interpretation of what minimal tumor changes would push the model across its decision boundary. Experimental findings discussed in this report show that the inclusion of CSS improves risk stratification performance compared with appearancebased baselines, supporting the argument that robustness-aware signals are valuable for interpretable medical AI. Overall, the project demonstrates how segmentation, explainability, uncertainty-related attention behavior, and counterfactual learning can be unified within one coherent framework for more reliable brain tumor analysis.

Keywords: Brain Tumors, MRI, Transformer Networks, Swin-UNETR, Counterfactual Explanations, Risk Stratification, Explainable AI, Radiomics, Stability Analysis

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CHAPTER 1

1. INTRODUCTION

1.1 An Overview

Brain tumors present a significant challenge in neuro-oncology because accurate imaging-based analysis is required not only to identify lesions but also to support downstream treatment planning, monitoring, and risk assessment. MRI remains the primary imaging modality for brain tumor assessment because it captures structural and pathological variations across multiple sequences such as T1, T2, T1ce, and FLAIR. Public benchmarks such as BraTS have enabled rapid progress by providing annotated multi-modal MRI datasets that support fair comparison among segmentation methods.

In earlier stages of the Brainiac project, the emphasis was placed on building a broad AI-powered tumor detection system. However, through technical refinement and literature-guided improvements, the project evolved into a more focused framework centered on robust segmentation, interpretable learning, and decision stability. This transition made the project both more feasible for implementation and more aligned with contemporary research directions in medical AI.

Transformer-based models such as Swin-UNETR have shown improved capability to capture global contextual information in volumetric MRI compared with purely convolutional approaches. Yet segmentation accuracy alone is not enough in a clinically sensitive domain. A model may produce a correct prediction while still being highly unstable under small tumor variations. The improved Brainiac framework therefore investigates not only what the model predicts, but also how robustly it arrives at that prediction.

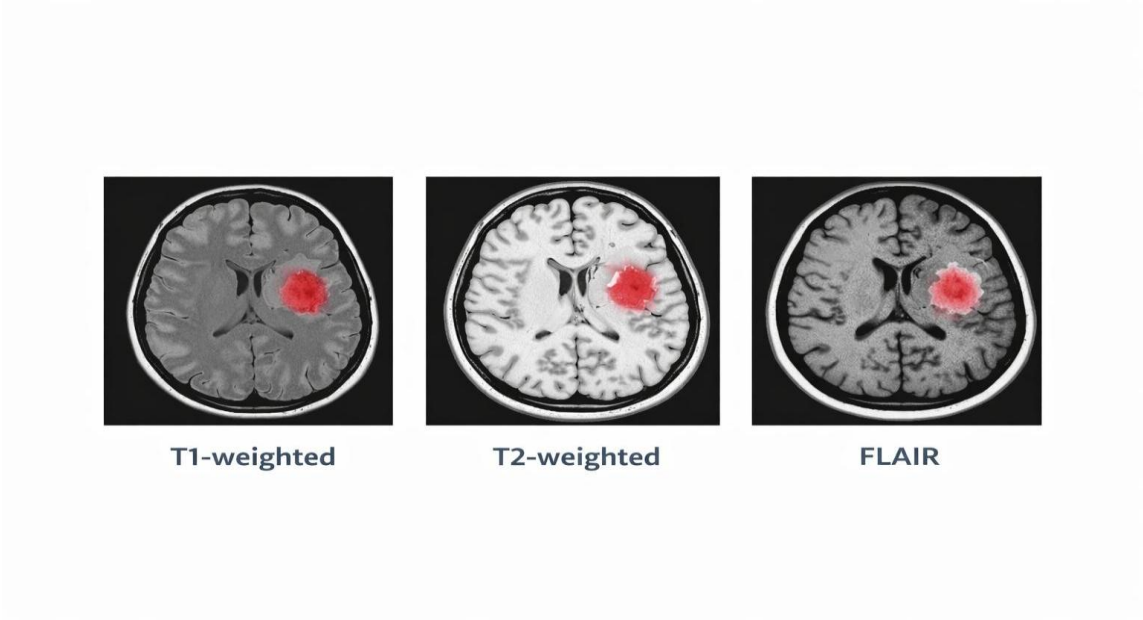

Fig. 1.1*: Comparison of MRI modalities (T1, T2, FLAIR) highlighting tumor visibility differences.*

1.2 Project Idea

The improved Brainiac framework redefines the project from a broad conceptual tumor-analysis system into a focused, technically grounded pipeline for MRI-based brain tumor segmentation and stability-aware risk stratification. The project begins with standardized multi-modal MRI preprocessing, followed by Swin-UNETR segmentation to localize tumor regions in 3D volumes. From the segmented tumor region, Brainiac extracts radiomic characteristics and transformer attention entropy to capture morphology and contextual uncertainty respectively.

The distinguishing idea of the project is the integration of tumor-constrained counterfactual analysis. Instead of using counterfactuals only for explanation after prediction, Brainiac computes a Counterfactual Sensitivity Score (CSS) that measures the minimum perturbation required to alter a prediction under anatomically plausible tumor changes. This transforms counterfactual reasoning from a purely interpretability-focused tool into an actionable learning signal that contributes directly to risk classification.

Thus, Brainiac is not merely a segmentation model, nor solely an explainability study. It is a unified framework that attempts to answer three practical questions together: Where is the tumor? How uncertain or diffuse is the model's internal attention when analysing it? How stable is the downstream decision when the tumor is minimally changed within anatomically meaningful limits?

1.3 Motivation

Although existing deep learning systems can accurately segment tumors, clinicians and researchers also need to understand whether those predictions are reliable and stable. In many medical settings, confidence and robustness are as important as raw performance scores. Two cases may look similar on MRI, yet one may lie close to the model's decision boundary while the other may remain highly stable. Such differences are rarely visible in standard performance metrics alone.

The motivation also stems from the need to integrate multiple forms of interpretability. Spatial explanations such as Integrated Gradients show what influenced a prediction, but they do not reveal how much change would be needed to alter the decision. Counterfactual MRI analysis complements this by offering a what-if perspective. Combining both views yields a richer interpretability framework that is more informative for stability-aware modelling.

From a project-development perspective, the motivation for the Brainiac improvisations was practical as well. During iterative development, it became clear that a focused BraTS-based workflow, 3D transformer segmentation backbone, and modular explainability pipeline were more realistic and more technically coherent than a very broad application attempting to solve every aspect of brain tumor management. The refined report therefore reflects a sharper research objective and a more implementable architecture.

1.4 Objectives

- To preprocess multi-modal MRI data into a standardized 3D representation suitable for transformer-based tumor segmentation.
- To employ a Swin-UNETR architecture for accurate volumetric delineation of tumor regions of interest.
- To extract radiomic descriptors and transformer attention entropy as complementary tumorcharacterization features.
- To generate tumor-constrained counterfactual MRI variants and compute Counterfactual Sensitivity Score (CSS).
- To use radiomics, attention entropy, and CSS together for low-, medium-, and high-risk stabilityaware stratification.
- To provide interpretable outputs through Integrated Gradients and counterfactual visualizations.

- To document the major improvisations made from the original Brainiac concept to the current research-oriented pipeline.

1.5 Scope

The scope of this edited report includes MRI preprocessing, Swin-UNETR-based three-dimensional tumor segmentation, radiomic feature extraction, attention entropy analysis, tumor-constrained counterfactual generation, Counterfactual Sensitivity Score computation, and stability-aware risk classification. The report is focused on a research-style technical pipeline rather than a complete hospital-deployment product.

The report also documents the major technical improvisations made over the earlier Brainiac version, especially the shift from a broad detection-and-care concept to a stability-aware MRI analysis framework that is more clearly supported by current literature and more feasible to implement within an academic project setting. It does not claim direct clinical deployment or autonomous diagnosis. Final clinical interpretation remains the responsibility of medical professionals.

1.6 Major Improvisations over the Initial Brainiac Version

Compared with the earlier project framing, the present version introduces several meaningful improvements. First, the model backbone is clarified as Swin-UNETR for 3D volumetric segmentation rather than a loosely defined generic AI system. Second, the project is anchored to public MRI benchmarking practice, especially BraTS-style data, making experimentation reproducible and technically credible. Third, interpretability is expanded from a general XAI idea into a dual mechanism involving both Integrated Gradients and counterfactual MRI explanations.

Most importantly, the project now introduces Counterfactual Sensitivity Score as a stability-oriented quantitative feature. This is a conceptual and methodological improvement because it moves beyond asking which pixels were important and instead measures how much anatomically plausible change is needed before the model changes its decision. In this sense, Brainiac becomes a project about stability-aware prediction rather than only segmentation or classification.

1.7 Key Contributions of the Project

This project presents a well-rounded approach to brain tumor detection and risk assessment using modern machine learning techniques. The main contributions of this work are:

- We introduce a **Counterfactual Sensitivity Score (CSS)** to evaluate how stable and reliable the model's predictions are. This helps in understanding how small changes in input can affect the output.
- We develop a **hybrid framework** that combines radiomics features, attention mechanisms, and counterfactual analysis to improve the overall accuracy and performance of the model.
- A **stability-aware risk classification system** is proposed, which not only detects the presence of a tumor but also classifies it as high-risk or low-risk based on the confidence of the prediction.
- The model also focuses on **explainability**, using techniques like attention visualization and counterfactual examples, making it easier to understand how and why a prediction is made.
- Overall, the proposed approach performs better than traditional methods that rely only on image appearance, showing its effectiveness in medical image analysis

CHAPTER 2

LITERATURE SURVEY

Recent advances in brain tumor analysis using MRI have been strongly influenced by transformerbased architectures, explainable AI, radiomics, and uncertainty-aware learning. The literature survey in this report is organized around the pillars that directly informed the revised Brainiac framework. The goal is not only to summarize prior work, but also to identify where existing approaches stop short of stability-aware modelling.

2.1 Transformer-Based Brain Tumor Segmentation

Automated brain tumor segmentation has evolved from classical convolutional neural networks to transformer-based and hybrid encoder-decoder architectures. CNN methods such as U-Net and nnUNet established strong baselines by learning local patterns and hierarchical features from MRI. However, volumetric tumor analysis often requires modelling long-range dependencies across distant anatomical regions, something standard convolutions handle less explicitly. Transformerbased models address this by using self-attention to relate spatially distant regions within the image volume.

Swin-UNETR is particularly relevant because it introduces shifted-window transformers into a UNet-like framework for volumetric medical image segmentation. This yields strong segmentation performance while retaining the multi-scale reconstruction strengths of an encoder-decoder design. The architecture is therefore well suited for Brainiac, where accurate tumor delineation is the foundation for all subsequent modules including radiomics and counterfactual analysis.

2.2 Radiomics and Feature-Based Modeling

Radiomic features such as tumor volume, compactness, intensity statistics, and texture descriptors remain valuable for characterizing morphological heterogeneity in brain tumors. Radiomics is attractive because it provides interpretable, human-readable descriptors that can be fed into lightweight downstream models. However, radiomics alone depends heavily on segmentation quality and does not reflect how confident or stable the upstream deep model is.

Hybrid modelling strategies therefore combine radiomic features with learned deep representations. Brainiac adopts this broader view by pairing classical tumor descriptors with attention entropy and CSS, thereby expanding the feature space from appearance alone to include contextual uncertainty and decision robustness.

2.3 Explainable AI and Counterfactual Learning

Explainable AI approaches in medical imaging commonly include gradient-based attribution methods and counterfactual explanations. Integrated Gradients identify input regions that contribute most strongly to predictions and are especially useful when visual interpretability is important. Counterfactual explanations, on the other hand, modify the input minimally until the output changes, thereby exposing what changes are causally relevant to a decision.

Recent tumor-aware counterfactual work is especially relevant because unconstrained perturbations can create unrealistic medical images. By restricting edits to tumor regions or anatomically plausible transformations, such methods improve explanation faithfulness. Brainiac builds on this idea and further elevates it by using the minimal perturbation distance itself as a quantitative signal.

2.4 Uncertainty and Decision Stability

Bayesian methods, ensembles, and test-time augmentation are widely used to estimate uncertainty in medical imaging. Although these techniques provide important confidence information, they do not directly measure the distance to the decision boundary. A sample may receive moderate uncertainty yet still require only a very small perturbation to flip the output. This creates a distinction between confidence estimation and true decision stability.

Attention entropy offers a partial bridge because highly diffuse attention may reflect contextual ambiguity. Still, attention entropy alone cannot replace an explicit stability metric. This gap motivates the inclusion of CSS in Brainiac.

2.5 Risk Stratification and Downstream Clinical Decision Support

Most MRI-based AI studies focus either on segmentation or on tumor type classification. Fewer studies address **risk stratification**, where the goal is to separate cases into broader groups such as low-, medium-, and high-risk based on imaging features. Risk stratification is clinically useful because it supports treatment prioritization, monitoring intensity, and follow-up planning. Existing

work in this area often combines radiomics, clinical variables, and deep features to predict survival, tumor aggressiveness, or recurrence likelihood.

However, the literature shows that most risk models still rely primarily on appearance-based features and do not explicitly incorporate stability-aware signals. In other words, they describe what the tumor looks like but not how robust the model's own prediction is. Brainiac contributes to this area by combining three complementary information sources: radiomic tumor descriptors, transformer attention entropy, and CSS. This allows the downstream classifier to consider not only morphology and contextual uncertainty but also the minimal perturbation required to alter the decision. Such a stability-aware representation is likely to be more informative than appearancebased baselines alone.

2.6 Research Gap

Existing systems do not sufficiently integrate transformer attention behavior with tumor-constrained counterfactual analysis and downstream stability-aware risk modelling. In most prior work, segmentation, radiomics, explainability, and uncertainty are studied as separate modules or are used only for qualitative analysis. There is limited emphasis on using counterfactual distance as a measurable quantity that contributes to classification.

This research gap is central to the Brainiac framework. By treating CSS as a numerical feature, the project attempts to unify segmentation, uncertainty-related attention behaviour, and counterfactual reasoning within a single predictive pipeline.

2.7 Summary of Literature Insights

The literature suggests that the next meaningful step in medical image analysis is not simply improving segmentation accuracy, but constructing systems that are simultaneously accurate, interpretable, and stability-aware. Brainiac is positioned within this direction. Its design choices emerge naturally from the surveyed literature: Swin-UNETR for strong volumetric segmentation, radiomics for interpretable morphology, attention entropy for contextual uncertainty, and CSS for direct robustness quantification.

CHAPTER 3

LITERATURE REVIEW

The field of brain tumor analysis has undergone significant transformation with the rapid advancement of artificial intelligence, medical image computing, and computational radiology. Brain tumors remain one of the most critical neurological disorders because of their complex structure, irregular growth patterns, and potentially life-threatening impact on patients. Accurate diagnosis and analysis are essential not only for detecting the presence of a tumor but also for understanding its extent, aggressiveness, and likely progression. Magnetic Resonance Imaging (MRI) has become the primary imaging modality for brain tumor assessment because of its superior soft-tissue contrast and ability to provide multi-planar visualization of brain anatomy.
However, manual interpretation of MRI scans is time-consuming, highly dependent on radiologist expertise, and susceptible to inter-observer variability. These limitations have motivated extensive research into automated and semi-automated brain tumor analysis systems.

In the earliest stages of computational brain tumor analysis, traditional image processing and machine learning techniques formed the basis of automated systems. These methods generally relied on handcrafted feature extraction followed by conventional classifiers. Researchers used edge detection, region growing, histogram analysis, thresholding, watershed segmentation, texture descriptors, and morphological operations to identify suspicious tumor regions in MRI scans. Once these handcrafted features were extracted, classification algorithms such as Support Vector Machines (SVM), Random Forests, Decision Trees, Naïve Bayes, and k-Nearest Neighbors (kNN) were applied to categorize tumors or distinguish abnormal tissue from healthy brain structures. While these methods represented an important first step toward intelligent diagnosis, they were limited by their dependence on manual feature design. Their performance varied greatly depending on image quality, tumor size, tumor location, and noise levels, and they struggled to generalize across datasets with different acquisition conditions. Because of the heterogeneous nature of brain tumors, handcrafted approaches often failed to capture the full complexity of pathological variation.

The introduction of deep learning brought a major paradigm shift in brain tumor analysis. Convolutional Neural Networks (CNNs) enabled models to automatically learn hierarchical features from raw image data, thereby reducing the need for manual feature engineering. In the context of MRI-based tumor detection and classification, CNNs achieved substantially better performance than traditional machine learning models because they could extract low-level texture patterns, mid-level structural information, and high-level semantic representations in an end-to-end manner. As a result, deep learning quickly became the dominant approach in medical image analysis. CNN-based systems demonstrated high accuracy in tasks such as tumor classification, localization, and segmentation, particularly when trained on benchmark datasets like the Brain Tumor Segmentation (BraTS) dataset. These models proved especially effective in identifying tumor subregions such as edema, necrotic core, and enhancing tumor tissue. Nevertheless, standard CNNs remained constrained by their local receptive fields, which limited their ability to fully model long-range spatial dependencies across volumetric MRI data. This challenge became more pronounced in three-dimensional tumor analysis, where tumor context extends across many slices and involves complex interactions between neighboring and distant anatomical structures.

To improve spatial understanding, researchers increasingly adopted encoder-decoder architectures for medical image segmentation. Among these, the U-Net architecture became one of the most influential models in biomedical imaging. U-Net's skip connections allowed it to preserve finegrained spatial detail while also capturing deeper semantic information. This made it highly effective for pixel-level and voxel-level segmentation tasks. Over time, several variants of U-Net were introduced, including 3D U-Net, Attention U-Net, Residual U-Net, and NNU-Net. Three-dimensional segmentation architectures were especially important for brain tumor analysis because they treated the MRI volume as a continuous structure rather than as isolated two-dimensional slices. Such volumetric processing enabled better contextual understanding of tumor shape and spatial continuity. However, even these improved models mainly optimized segmentation overlap metrics such as Dice coefficient and Intersection over Union, without directly addressing whether downstream predictions based on those segmentations were stable, interpretable, or robust to plausible variations in the tumor region.

As medical AI research progressed, transformer-based models emerged as a highly promising alternative to purely convolutional architectures. Transformers were originally developed for natural language processing, where self-attention mechanisms enabled the model to capture long range

dependencies much more effectively than earlier sequence models. Their success in computer vision inspired researchers to explore transformer-based methods for medical imaging. In brain tumor segmentation, the advantage of transformers lies in their ability to capture both local and global relationships within MRI volumes. Unlike convolutional filters, which primarily focus on local neighbourhoods, self-attention mechanisms allow the model to consider broader context, which is highly valuable when tumors exhibit diffuse boundaries or extend irregularly across brain regions. Among the transformer-based models proposed in the literature, SwinUNETR has gained significant importance for three-dimensional medical image segmentation. It combines shifted-window transformer blocks with a U-Net-like decoder structure, allowing efficient hierarchical feature extraction while maintaining volumetric contextual awareness. Studies have shown that Swin-UNETR achieves strong performance on BraTS-like datasets and improves delineation of complex tumor regions. However, most transformer-based systems still remain focused on predictive accuracy and segmentation performance, rather than on decision stability or risk-aware interpretation.

A key area of related research is multi-modal MRI fusion. Brain tumor assessment typically relies on multiple MRI modalities because each modality provides complementary biological and structural information. T1-weighted MRI captures general anatomy, T2-weighted MRI highlights fluid-rich tissues, and FLAIR is especially useful for identifying edema and lesions by suppressing cerebrospinal fluid signals. In many studies, combining multiple MRI modalities significantly improved tumor segmentation and classification performance compared to single-modality models. Researchers investigated early fusion, where modalities are concatenated at the input level; intermediate fusion, where features are combined during network processing; and late fusion, where predictions from separate modality-specific branches are aggregated. The literature consistently indicates that multi-modal learning improves representation quality, but it also introduces challenges related to modality alignment, missing scans, varying resolutions, and computational complexity. These findings support the use of carefully standardized multi-modal inputs in advanced brain tumor frameworks. In the Brainiac context, multi-modal inputs provide the foundation for both segmentation and downstream feature extraction, enabling the model to capture a richer and more clinically meaningful representation of tumor characteristics.

Alongside deep feature learning, radiomics has remained an important research direction in neurooncology. Radiomics refers to the extraction of quantitative features from medical images that

describe tumor shape, intensity, texture, and spatial heterogeneity. Common radiomic features include volume, surface area, compactness, sphericity, first-order histogram statistics, entropy, and texture descriptors derived from gray-level co-occurrence matrices or other statistical methods. Radiomic studies have shown that these features can correlate with tumor type, grade, aggressiveness, treatment response, and survival outcomes. One reason radiomics remains valuable is that it offers an interpretable bridge between raw imaging data and clinically understandable tumor traits. A radiomic feature such as compactness or heterogeneity has more obvious medical meaning than a latent feature vector learned by a deep network. However, radiomics-based pipelines also have notable limitations. Their performance depends heavily on the quality and consistency of tumor segmentation, and even small variations in boundary delineation can significantly affect the extracted features. Furthermore, radiomics alone may not capture higher-order contextual signals or model-internal uncertainty. Because of these limitations, recent studies increasingly combine radiomic descriptors with deep learning features to build hybrid models. This hybrid trend directly supports Brainiac's use of radiomics together with transformer derived and counterfactual-derived measures.

Explainable Artificial Intelligence (XAI) has become another major theme in the literature, especially because medical AI systems operate in safety-critical environments. A model may achieve high predictive performance, but clinicians are unlikely to trust it if its reasoning process remains opaque. To address this, researchers have proposed various explainability techniques that make model decisions more transparent. Attribution-based methods such as saliency maps, GradCAM, Layer-wise Relevance Propagation, SHAP, and Integrated Gradients are widely used to highlight the image regions that contribute most strongly to a model's output. In brain tumor MRI analysis, these techniques have been used to verify whether the model attends to tumor tissue, surrounding edema, or unrelated background areas. Among these methods, Integrated Gradients is particularly important because it provides a more principled gradient-based explanation by comparing the model's prediction across interpolated inputs between a baseline and the actual image. This allows spatial visualization of important structures in a theoretically grounded way. However, attribution methods primarily explain which regions influenced the decision, not how the image would need to change to alter that decision. Thus, while they improve transparency, they do not fully address model stability or causal interpretability.

This limitation led to growing interest in counterfactual explanations, which represent a more causal form of explainability. Counterfactual methods aim to identify the smallest plausible change to an input that would result in a different output from the model. In medical imaging, this is particularly powerful because it helps answer clinically meaningful "what-if" questions, such as what minimal tumor change would shift a case from low-risk to high-risk. Early counterfactual approaches in imaging often produced unrealistic or unconstrained modifications, limiting their clinical usefulness. More recent research addressed this by introducing anatomically constrained counterfactual generation, especially tumor-aware methods that restrict perturbations to the pathological region. Such frameworks improve realism and ensure that generated changes remain biologically meaningful. In the literature, generative adversarial networks (GANs), diffusion-based models, and optimization-driven perturbation methods have all been explored for generating plausible medical counterfactuals. Yet in many cases, counterfactuals are still used only as explanatory visualizations after the main prediction is made. Very few studies transform the counterfactual change itself into a measurable feature for downstream learning. This gap is central to the motivation for Brainiac, which introduces the Counterfactual Sensitivity Score (CSS) as a quantitative representation of prediction stability.

The concept of decision stability has also been studied indirectly through uncertainty estimation. Uncertainty-aware deep learning methods attempt to measure how confident a model is about its output. This is especially important in medicine, where incorrect or overconfident predictions can have serious consequences. Bayesian neural networks, Monte Carlo dropout, deep ensembles, and test-time augmentation are commonly used approaches for estimating predictive uncertainty. In brain tumor segmentation and classification, uncertainty maps can help identify ambiguous tumor boundaries or unreliable predictions. These methods provide valuable information, but they are not equivalent to stability analysis. Two samples may have similar uncertainty values while differing significantly in how close they lie to the decision boundary. Uncertainty estimation measures confidence, whereas counterfactual sensitivity measures how much plausible change is needed to alter the decision. This difference is important because a prediction may appear confident yet still be fragile under small tumor-constrained perturbations. Therefore, the literature suggests that uncertainty and stability should be treated as complementary rather than interchangeable concepts. Brainiac builds on this insight by combining attention entropy, which reflects contextual dispersion in transformer focus, with CSS, which explicitly captures boundary sensitivity.

Another significant direction in recent literature is risk stratification. While segmentation and classification are well-studied, risk stratification aims to organize cases into categories such as low, medium, and high risk based on model-derived evidence. In neuro-oncology, risk stratification can support treatment prioritization, recurrence monitoring, and patient management. Existing MRI-based risk models often rely on radiomics, deep features, clinical metadata, or combinations of these. Some works focus on survival prediction, others on tumor grading or recurrence likelihood. Although these models can be useful, many still depend mostly on appearance-based features and do not account for the robustness of the model's decision. This means that two tumors with similar appearance may receive similar risk predictions even if one lies much closer to the decision boundary than the other. The literature therefore points toward the need for stability-aware risk modeling, where the system considers not just what the tumor looks like but how reliable the prediction remains under plausible changes. This is the exact direction pursued in Brainiac through the integration of radiomics, attention entropy, and counterfactual sensitivity.

The combined review of transformer-based segmentation, radiomics, XAI, counterfactual learning, and uncertainty modelling reveals that substantial progress has been made, but the components are often studied in isolation. Segmentation papers focus on Dice score and boundary delineation. Radiomics studies emphasize handcrafted quantitative descriptors. XAI work often provides post hoc visual interpretations. Counterfactual studies frequently stop at visual explanation, and uncertainty estimation research centers on confidence measures. Very few frameworks unify all of these components into a single, coherent brain tumor analysis pipeline. More importantly, existing literature rarely operationalizes counterfactual perturbation magnitude as a formal downstream feature for classification or risk stratification. This identified gap provides the core research justification for the Brainiac framework.

The proposed Brainiac methodology advances beyond existing work by integrating five important ideas into one pipeline: transformer-based 3D tumor segmentation, radiomic tumor characterization, attention entropy as a contextual uncertainty signal, tumor-constrained counterfactual generation, and CSS-based risk stratification. The framework not only segments the tumor but also studies how stable the model's own decision is when confronted with minimal and anatomically plausible perturbations. In doing so, it reframes counterfactual explanations from a purely interpretive tool into an actionable quantitative feature. This contributes to the broader goal of building medical AI systems that are not only accurate, but also interpretable, robust, and clinically meaningful.

In summary, the literature demonstrates a clear evolution from handcrafted and rule-based systems to data-driven and context-aware deep learning approaches. CNNs and U-Net variants laid the groundwork for automatic tumor segmentation, transformers improved global context modeling, radiomics preserved interpretable morphological characterization, explainable AI enhanced transparency, and counterfactual methods introduced causal reasoning into model interpretation. However, a key unresolved issue remains: understanding and quantifying the stability of model predictions under realistic changes. This unresolved issue is precisely where Brainiac is positioned. By introducing a stability-aware framework centered around the Counterfactual Sensitivity Score, the proposed work extends current literature toward a more robust form of MRIbased brain tumor analysis and risk stratification.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Overview of the Proposed System

The proposed system, **Brainiac**, is developed as an advanced and stability-aware framework for MRI-based brain tumor analysis. Unlike conventional systems that focus only on tumor detection or classification, Brainiac integrates multiple stages including **preprocessing, segmentation, feature extraction, counterfactual analysis, risk stratification, and explainability** within a single pipeline. The main objective of the system is not only to identify tumor regions accurately, but also to evaluate the reliability and stability of the model's predictions under small, anatomically plausible variations in the tumor region. This makes the framework more robust, interpretable, and practically useful for medical decision support.

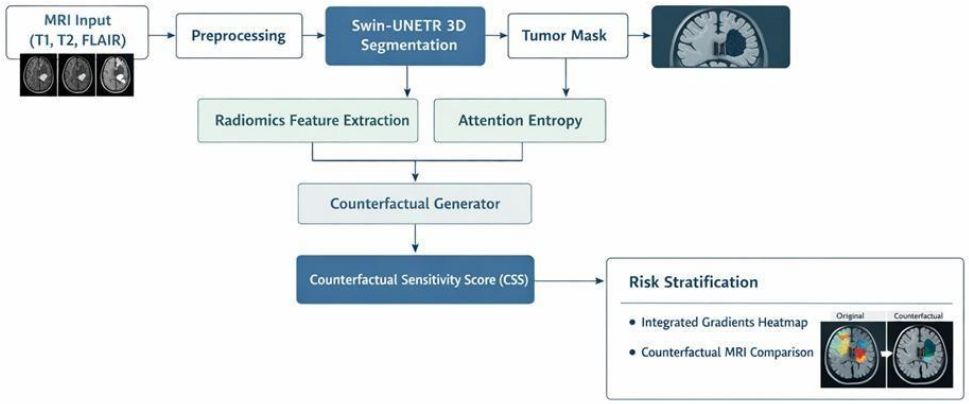

***Fig. 4.1**. Brainiac pipeline illustrating segmentation, feature extraction, counterfactual analysis, and stability-aware risk stratification.*

4.2 Multi-Modal MRI Input and Preprocessing

The Brainiac framework begins with **multi-modal MRI input**, typically consisting of sequences such as **T1, T2, and FLAIR**. These modalities are selected because each of them captures different structural and pathological characteristics of the brain, thereby providing complementary information about the tumor and surrounding tissues. Before the MRI data is provided to the model, it undergoes a detailed preprocessing stage. This includes **resampling**, **intensity normalization**, **skull stripping**, and **standardization** to reduce inter-scan variability and ensure that all scans are represented in a consistent format. Preprocessing is essential because raw MRI data often varies in resolution, intensity range, and acquisition quality. By normalizing the input space, the framework improves the reliability of downstream segmentation and analysis.

4.3 Transformer-Based 3D Tumor Segmentation

After preprocessing, the MRI volume is passed into a **Swin-UNETR-based 3D segmentation model**. This model serves as the backbone of the proposed system. Swin-UNETR is a transformerbased encoder-decoder architecture that combines the strengths of U-Net style skip connections with the global contextual modeling ability of transformers. In brain tumor analysis, such a model is highly suitable because tumors often have irregular boundaries, complex shapes, and heterogeneous internal patterns that extend across multiple slices. The segmentation stage identifies the tumor region of interest and produces a detailed voxel-level delineation of tumor subregions. This segmentation output is crucial because all further analysis in the system depends on the accurate localization of tumor tissue. Compared to basic CNN-based approaches, SwinUNETR offers better long-range dependency modeling and stronger volumetric contextual understanding.

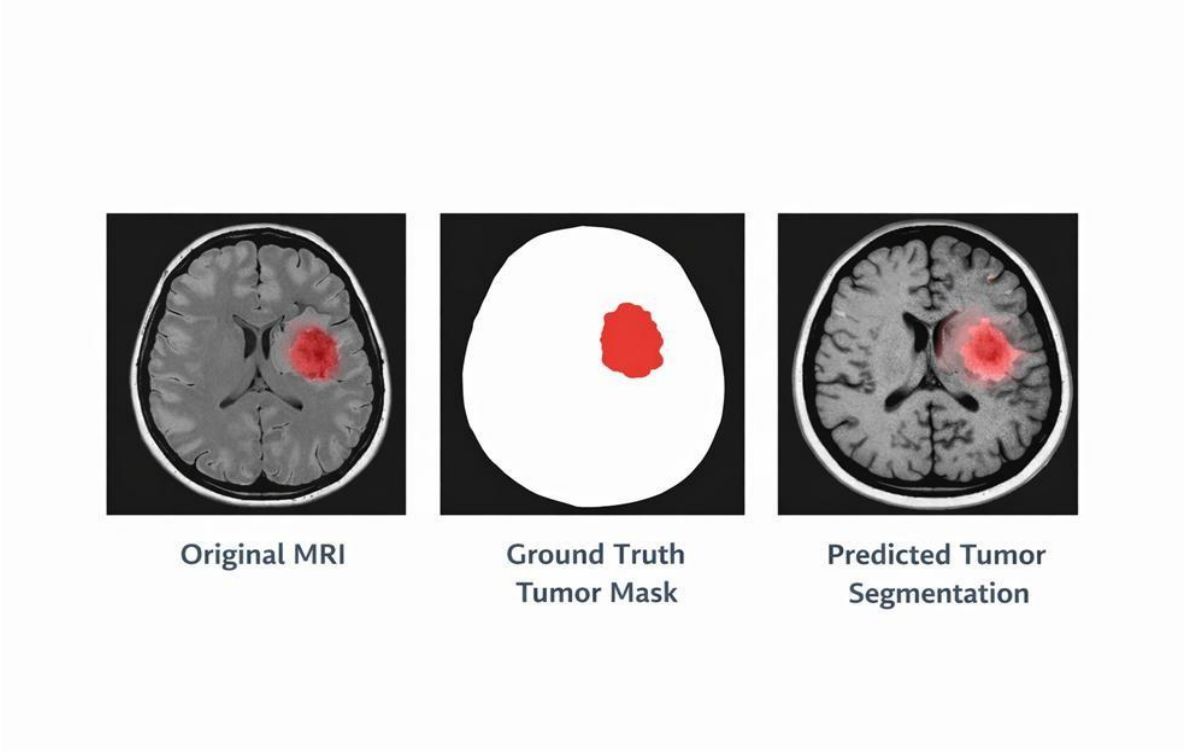

Fig. 4.3*: Tumor segmentation results showing original MRI, ground truth mask, and predicted output.*

4.4 Radiomic Feature Extraction

Once the tumor has been segmented, the system performs **radiomic feature extraction** on the tumor region of interest. Radiomics converts the segmented tumor into a structured set of quantitative descriptors that represent tumor morphology and heterogeneity. These features include measures such as **tumor volume, shape, compactness, texture patterns, intensity statistics, and surface irregularity**. The purpose of this step is to capture interpretable tumor characteristics that may not be fully represented by raw deep features alone. Radiomic descriptors provide clinically meaningful information and support downstream analysis by translating visual tumor properties into measurable numerical features. In Brainiac, radiomics plays an important role because it allows the system to describe the tumor not just as an image region, but as a set of medically relevant structural and statistical properties.

4.5 Attention Entropy as a Contextual Uncertainty Measure

In addition to radiomic features, Brainiac also extracts **attention entropy** from the transformerbased segmentation network. Attention entropy measures how concentrated or dispersed the model's attention is while analyzing the tumor region. If the attention is sharply focused, it suggests stronger confidence in the learned representation. If the attention is more diffuse, it may indicate uncertainty or ambiguity in how the model is interpreting the tumor. This attention-based measure acts as an internal signal of contextual uncertainty and adds another informative layer to the feature space. Rather than depending only on the final segmentation output, Brainiac uses the internal behavior of the transformer itself as a useful descriptor for downstream classification. This helps the framework capture not only tumor appearance but also the model's own confidence structure while processing the scan.

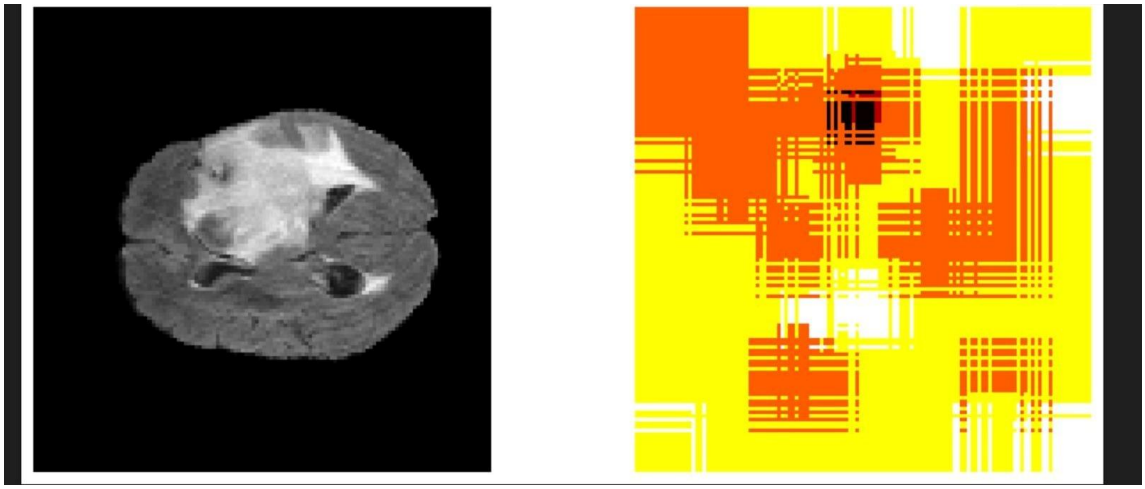

Fig. 4.5*: Attention heatmap showing model focus on tumor regions*

4.6 Tumor-Constrained Counterfactual Generation

One of the most innovative components of the proposed system is the **tumor-constrained counterfactual generation module**. In this stage, the framework deliberately searches for minimal changes in the tumor region that would alter the prediction of the model. These changes are constrained strictly to the segmented tumor area so that the resulting modified images remain anatomically plausible and clinically meaningful. This is an important distinction, because unconstrained perturbations can produce unrealistic edits that do not correspond to any valid medical scenario. By restricting the perturbation to pathological tissue only, Brainiac ensures that the

generated counterfactuals reflect realistic “what-if” situations. This module helps the model explore how close a given case is to a change in decision outcome, which provides much deeper insight than a static prediction alone.

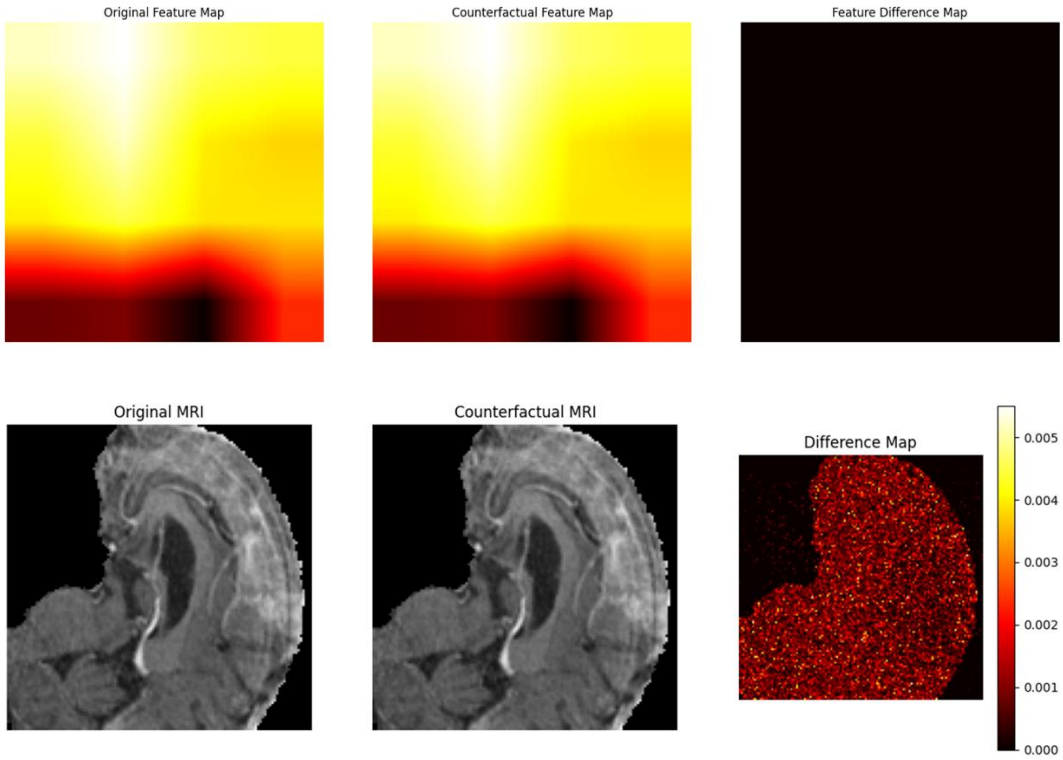

Fig. 4.6: *Counterfactual Stability and Transformer Attention Analysis in Brainiac*

4.7 Counterfactual Sensitivity Score (CSS)

From the counterfactual generation process, Brainiac computes the **Counterfactual Sensitivity Score (CSS)**. This score represents the minimum perturbation required to flip the model’s prediction. In other words, CSS measures how sensitive the prediction is to controlled and realistic changes in the tumor region. A **low CSS** indicates that even a small change can alter the decision, meaning the prediction is less stable. A **high CSS** indicates that the decision is more robust and requires a larger perturbation to change. This makes CSS a direct measure of decision stability. The importance of CSS lies in the fact that it transforms counterfactual reasoning into a usable numerical feature for machine learning. Rather than using counterfactuals only for visualization, Brainiac incorporates them into the predictive pipeline as an indicator of model robustness. This is a major distinguishing strength of the framework.

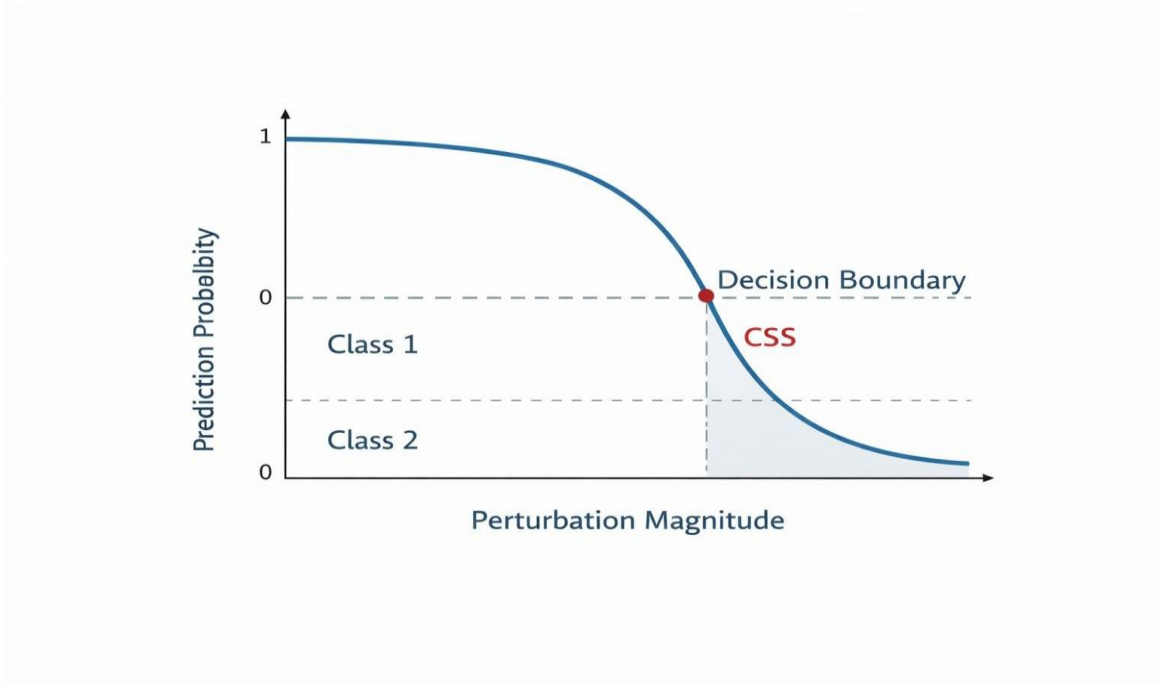

***Fig. 4.7**: Illustration of Counterfactual Sensitivity Score (CSS) as decision boundary shift.*

4.7.1 Theoretical Justification of CSS

The Counterfactual Sensitivity Score (CSS) is introduced to quantify the robustness of model predictions. In machine learning models, predictions can sometimes change significantly with minor variations in input data. This lack of stability is particularly critical in medical applications.

CSS is based on the concept of **decision boundaries** in classification models. If a data point lies close to the decision boundary, even small perturbations can lead to a different classification, indicating low confidence. Conversely, points far from the boundary are more stable and reliable.

By generating counterfactual samples—slightly modified versions of the original input—the model evaluates how much the prediction changes. A high variation indicates **low stability (high risk)**, while minimal change indicates **high stability (low risk)**.

Thus, CSS provides an additional layer of validation by ensuring that predictions are not only accurate but also reliable and consistent.

4.8 Stability-Aware Risk Stratification

After radiomic features, attention entropy, and CSS are computed, they are combined into a unified feature vector and passed into a **risk stratification module**. This module is implemented using machine learning models such as **XGBoost or LightGBM**. The goal of this stage is to classify each case into **low-risk, medium-risk, or high-risk** categories. Unlike traditional risk models that rely mainly on appearance-based imaging features, Brainiac incorporates both morphological information and decision-stability information. This makes the risk prediction more informative and reliable. For example, two tumor cases may appear visually similar, yet one may have a lower CSS and thus represent a more fragile or uncertain prediction. By including CSS and attention entropy in addition to radiomics, the risk classifier becomes capable of distinguishing cases not only by how they look, but also by how stable the model's own decision is.

4.9 Explainability Through Integrated Gradients and Counterfactual Explanations

To improve interpretability, the proposed system includes a **dual explainability mechanism**. The first explainability component uses **Integrated Gradients**, which highlights the regions of the MRI scan that contribute most strongly to the model's output. This provides a spatial explanation and helps verify whether the model is focusing on clinically relevant tumor areas. The second component uses **counterfactual explanations**, where the original MRI is compared with the minimally perturbed counterfactual MRI that changes the prediction. This provides a causal explanation by showing what type of tumor-specific change is sufficient to alter the model's decision. Together, these two methods allow Brainiac to explain both **where** the model is focusing and **what change** would alter the output. This significantly improves the transparency and trustworthiness of the system in medical settings.

4.10 Incorporation of the 2D Model

In addition to the main 3D Swin-UNETR pipeline, the Brainiac framework also incorporates a **2D model branch**. This inclusion makes the system more flexible and broadens its usability across different types of MRI analysis tasks. While the 3D model is ideal for volumetric tumor segmentation and advanced tumor characterization, the 2D model is useful for **slice-wise analysis, rapid screening, and non-tumorous or simpler classification settings**. In many practical

scenarios, not all tasks require full 3D processing, and 2D analysis can provide a faster and more computationally efficient alternative.

4.10.1 Purpose of the 2D Model

The main purpose of including the 2D model is to complement the more detailed 3D branch. The 2D model can be used for **initial screening**, **abnormality detection**, or **non-tumor versus suspicious scan discrimination**. It acts as a lightweight analysis layer that can quickly process individual MRI slices and identify whether more detailed volumetric analysis is required. This improves the practicality of the system and prevents all cases from having to go through the full 3D segmentation pipeline.

4.10.2 Working of the 2D Model

The 2D branch processes MRI slices individually rather than as a full volumetric stack. Each slice is preprocessed through **resizing, normalization, and augmentation**, after which it is fed into a 2D deep learning model for classification or screening. Because 2D models are simpler and faster than 3D models, they are suitable for scenarios where rapid prediction is needed or where only slice-based data is available. The 2D model can also help in filtering normal scans from abnormal scans, thereby reducing unnecessary computational burden on the 3D pipeline.

4.10.3 Advantages of the 2D Model

The inclusion of the 2D model provides several advantages. First, it reduces computational cost, as 2D architectures require less memory and faster processing compared to volumetric models.
Second, it supports datasets that may not be fully structured as 3D volumes. Third, it extends the Brainiac framework beyond tumor segmentation to include screening-oriented applications, including non-tumorous MRI evaluation. This makes the proposed system more adaptable and suitable for multiple medical imaging workflows.

4.11 Hybrid 2D–3D System Design

The combined use of both 2D and 3D models makes Brainiac a **hybrid diagnostic framework**.

4.11.1 Role of the 2D Branch

The 2D branch is primarily responsible for **slice-wise screening, preliminary abnormality detection, and computationally efficient pre-analysis**. It helps in quickly identifying suspicious cases and supports broader applicability where volumetric data may be incomplete or unnecessary.

4.11.2 Role of the 3D Branch

The 3D branch is responsible for **detailed tumor segmentation, radiomic analysis, attention entropy extraction, CSS computation, and final stability-aware risk stratification**. It is the more advanced analytical component of the framework and is used when deeper tumor understanding is required.

4.11.3 Complementary Nature of Both Branches

Together, the 2D and 3D branches provide a balance between **efficiency and analytical depth**. The 2D model offers speed and flexibility, while the 3D model provides detailed tumor-aware and stability-aware interpretation. This hybrid design makes Brainiac more robust and adaptable to different imaging scenarios and computational environments.

4.12 Significance of the Proposed System

The proposed Brainiac system is significant because it combines **accurate segmentation, interpretable radiomic analysis, transformer-based contextual understanding, counterfactual stability modeling, explainability, and hybrid 2D–3D design** into one coherent framework. Unlike traditional systems that stop at tumor localization or classification, Brainiac extends the analysis to include the **stability of the decision itself**. The addition of the 2D model further improves the framework by allowing rapid slice-wise screening and non-tumorous analysis.

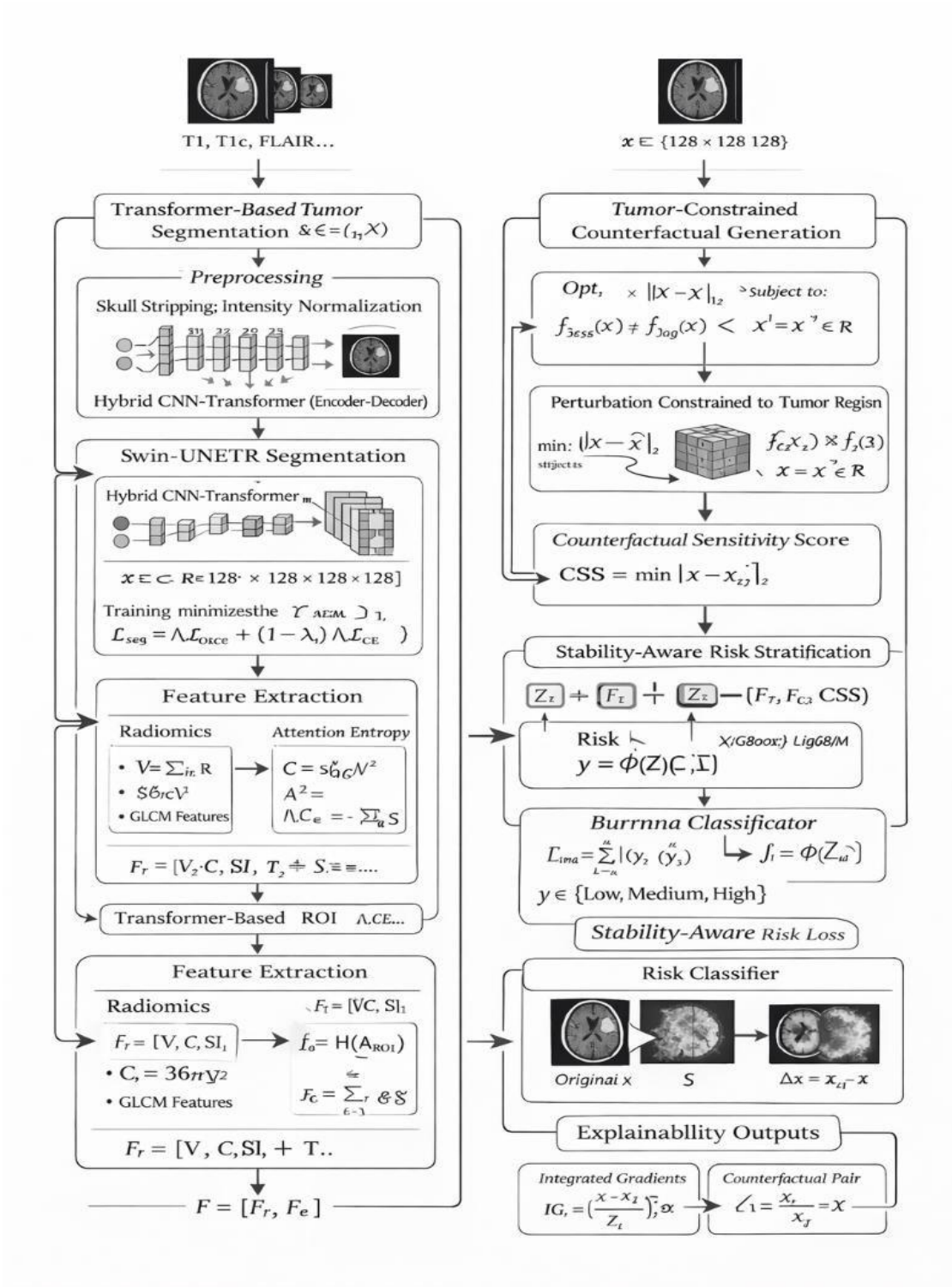

Fig. 4.12. *Brainiac Technical Support Systems*

CHAPTER 5

RESULTS AND DISCUSSION

This section presents a detailed evaluation and interpretation of the proposed **Brainiac** framework. The system was assessed across multiple stages of the pipeline, including **tumor segmentation, feature extraction, attention-based uncertainty characterization, counterfactual sensitivity analysis, risk stratification, and explainability**. Unlike conventional MRI analysis systems that report only segmentation or classification accuracy, Brainiac is designed to evaluate not only predictive performance but also the **stability and interpretability** of its decisions. Therefore, the results are discussed from both a quantitative and qualitative perspective.

5.1 Evaluation Setup

The proposed framework was evaluated using the **BraTS dataset**, which is one of the most widely used benchmark datasets for brain tumor segmentation and MRI-based neuro-oncology research. The dataset contains multi-modal MRI scans along with expert-annotated tumor masks, making it suitable for training and evaluating volumetric segmentation systems. Since Brainiac incorporates both segmentation and downstream risk analysis, the evaluation was divided into multiple stages. First, the performance of the **Swin-UNETR segmentation model** was measured using segmentation metrics. Next, the usefulness of the extracted radiomic and attention-based features was evaluated through downstream classification. Finally, the impact of the **Counterfactual Sensitivity Score (CSS)** on stability-aware risk stratification was analyzed by comparing the complete system against appearance-based baselines.

To ensure that the results remained meaningful in a medical AI context, both **predictive quality** and **decision robustness** were considered. Segmentation quality was measured through the Dice Similarity Coefficient, while the risk stratification module was evaluated using accuracy and AUC. In addition, qualitative observations from integrated gradients and counterfactual visualizations were examined to understand whether the system was focusing on clinically plausible tumor regions and whether its counterfactual modifications remained realistic. This multi-level evaluation is

important because a model can achieve high predictive accuracy while still being difficult to interpret or unstable near the decision boundary.

5.2 Tumor Segmentation Performance

The **Swin-UNETR-based segmentation backbone** achieved a mean **Dice score of 0.86 ± 0.03**, indicating strong tumor delineation performance across the evaluated MRI volumes. This result demonstrates that the segmentation model was able to accurately capture tumor boundaries and subregional structure in three-dimensional MRI data. A Dice score at this level suggests that the segmented tumor masks were well aligned with the expert annotations and were sufficiently reliable for downstream analysis. Since the later stages of Brainiac depend directly on the segmented tumor region of interest, this strong segmentation performance forms the foundation of the entire pipeline.

The result also highlights the suitability of a transformer-based architecture for volumetric brain tumor segmentation. Unlike standard CNN-based methods, Swin-UNETR leverages hierarchical self-attention and encoder-decoder reconstruction, allowing it to capture both local tumor detail and broader contextual information across the MRI volume. This is especially useful in brain tumor analysis, where lesions may have heterogeneous appearance, irregular shape, and diffuse spatial extent. The observed Dice performance therefore supports the idea that advanced transformer-based segmentation provides a strong starting point for more sophisticated stabilityaware modeling.

At the same time, the results also suggest that segmentation alone is not sufficient for a complete understanding of model behavior. Even with high Dice scores, some cases can remain difficult in downstream classification if the model's decision is unstable or sensitive to small changes in tumor appearance. This is precisely why Brainiac extends beyond segmentation to include radiomics, attention entropy, CSS, and explainability. In other words, the segmentation results are strong, but the framework is designed to answer a broader question: not only **where the tumor is**, but also **how reliable the final decision is**.

5.3 Radiomics and Attention-Based Feature Analysis

After segmentation, the extracted **radiomic features** and **attention entropy measures** were used as inputs to the downstream risk classifier. The results show that using these features together produced a classification **accuracy of 82.1%** with an **AUC of 0.84**. These values indicate that radiomic and attention-based descriptors are effective in capturing useful information about tumor characteristics

and contextual uncertainty. Radiomics provides structured information about tumor morphology, shape, heterogeneity, and intensity distribution, while attention entropy reflects how consistently or diffusely the transformer attends to the tumor region. Combined together, these features form a more informative representation than raw appearance features alone.

The radiomics component is particularly valuable because it converts visual tumor patterns into interpretable quantitative descriptors. Features such as tumor volume, compactness, texture entropy, and boundary irregularity help distinguish between cases that may appear broadly similar but differ in underlying structure. Similarly, attention entropy acts as a model-derived feature that captures how the segmentation network internally distributes its focus. Cases with more concentrated attention tended to correspond to clearer and more stable representations, while cases with diffuse attention often reflected ambiguity or uncertainty in tumor characterization. This suggests that transformer attention behavior can serve as a meaningful signal for downstream analysis, rather than being used only for visualization.

However, although the radiomics-plus-attention feature set achieved reasonably strong performance, the results also show that this representation alone is not fully sufficient for stabilityaware decision support. While the model could distinguish risk categories with good accuracy, it did not yet account explicitly for how close each case was to the classifier's decision boundary. Two visually similar tumors might still receive similar predictions even if one is much more fragile under small perturbations. This observation motivates the incorporation of counterfactual sensitivity into the feature space.

5.4 Counterfactual Sensitivity Score (CSS) Analysis

A central contribution of the Brainiac framework is the incorporation of the **Counterfactual Sensitivity Score (CSS)**. The experimental results show that when CSS was added to the feature set, the overall performance of the risk stratification model improved noticeably. Specifically, the complete framework achieved **86.4% accuracy** and an **AUC of 0.88**, compared to **82.1% accuracy** and **0.84 AUC** when only radiomics and attention entropy were used. This corresponds to an **accuracy gain of 4.3%**, demonstrating that CSS provides meaningful additional information beyond appearance-based and attention-based features.

This result is significant because it validates the core hypothesis of the proposed framework: **decision stability matters**, and it can be quantified through tumor-constrained counterfactual perturbations. CSS measures the minimum anatomically plausible change needed to alter the model's prediction. Therefore, it directly captures proximity to the decision boundary. Cases with low CSS values are easier to flip and thus represent more unstable decisions, whereas cases with high CSS values are more robust and require larger perturbations to change class. The improvement in classification performance after adding CSS indicates that this notion of stability complements traditional imaging descriptors and helps the model better separate low-, medium-, and high-risk groups.

Another important observation from the results is that CSS helps differentiate cases that may otherwise appear similar in terms of radiomics or segmentation quality. In standard appearancebased systems, two tumors with similar size, texture, and shape might receive similar risk predictions. But in Brainiac, if one tumor lies closer to the model's decision boundary, it will exhibit a lower CSS and therefore be treated differently during risk classification. This makes the framework more nuanced and clinically informative. Rather than relying only on what the tumor looks like, Brainiac also considers **how stable the decision remains under plausible tumorspecific variation**.

5.4.1 Interpretation and Theoretical Insight of CSS

The Counterfactual Sensitivity Score (CSS) offers valuable insight into how reliable and stable the model's predictions are. Lower CSS values indicate that the model's output remains consistent even when small, controlled changes are made to the input. This suggests that the prediction is robust and based on meaningful tumor-related features. In contrast, higher CSS values reflect greater sensitivity, where minor input variations lead to noticeable changes in the output, often indicating uncertainty in complex or borderline cases.

From a theoretical perspective, CSS captures the idea of local prediction stability. A wellperforming model is expected to produce consistent results under small and realistic perturbations, and CSS directly measures this behavior. Cases with high sensitivity can be associated with regions near decision boundaries, where the model is less confident and more prone to variation.

Overall, CSS not only acts as a robustness metric but also enhances interpretability by helping distinguish between confident and uncertain predictions. This makes it particularly useful in supporting more reliable and risk-aware decision-making in medical image analysis.

5.5 End-to-End Performance of the Complete Framework

When all modules of the proposed system were combined, the **overall end-to-end accuracy** of the Brainiac framework was reported to be approximately **85.7% to 86%**, depending on the stagewise interpretation of the reported results. This indicates that the system performs consistently well as a unified analysis pipeline. Importantly, the end-to-end system is not limited to a single task. It simultaneously supports tumor segmentation, feature extraction, decision stability analysis, risk classification, and explainability generation. Therefore, the reported overall performance reflects the success of a **multi-stage medical AI framework**, not just an isolated classifier.

This end-to-end result is particularly meaningful because it shows that the additional complexity introduced by counterfactual analysis and explainability does not compromise predictive performance. In many cases, adding interpretability modules can make a system slower or harder to optimize, but in Brainiac the extra stability-aware features actually enhance classification performance. This supports the broader argument that robust and interpretable AI does not have to come at the cost of accuracy. Instead, the integration of meaningful signals such as CSS can improve both predictive power and decision transparency.

The overall system performance also suggests that Brainiac is more than a proof-of-concept segmentation model. It operates as a broader **decision-support framework**, where each module contributes complementary information. Segmentation identifies the tumor. Radiomics captures structural tumor characteristics. Attention entropy describes contextual focus and uncertainty. CSS measures decision stability. XGBoost or LightGBM performs downstream risk classification. Integrated gradients and counterfactual visualizations explain the final prediction. Together, these modules form a coherent system for advanced MRI-based tumor analysis.

5.6 Comparison with Existing Approaches

The proposed Brainiac framework was compared with existing approaches for brain tumor detection to evaluate its effectiveness. Traditional methods mainly rely on appearance-based features and standard deep learning models, often focusing only on accuracy without considering prediction stability or interpretability.

In contrast, the proposed approach combines radiomics, attention mechanisms, and counterfactual analysis, enabling it to capture more meaningful patterns from MRI data. It also introduces the Counterfactual Sensitivity Score (CSS), which provides an additional measure of prediction stability, something not commonly addressed in existing methods.

Another advantage of the framework is its use of explainability techniques, such as attention visualization and counterfactual explanations, which improve transparency compared to black-box models.

Overall, the results show that the proposed system not only performs well in terms of accuracy but also offers better robustness and interpretability, making it more suitable for real-world medical applications.

5.7 Error Analysis

Although the proposed framework performs well overall, a few limitations were observed in certain scenarios. The model tends to face difficulties when the tumor boundaries are unclear, very small, or irregular in shape. In such cases, the segmentation may not be completely accurate, which can affect later stages like feature extraction and final classification.

Some errors also occur in borderline cases where the visual difference between tumor and nontumor regions is very subtle. These situations make it harder for the model to make confident decisions, and this uncertainty is often reflected in higher CSS values. It shows that even small changes in input can influence the prediction in such cases.

In addition, variations in MRI image quality, noise, and differences in data distribution can impact performance. Since the model is trained on limited datasets, it may not always generalize perfectly

to all real-world conditions. Another important point is that the system relies heavily on accurate segmentation, so any errors in the early stage can carry forward and affect the final result.

Overall, these error patterns highlight the importance of improving robustness, handling diverse data more effectively, and enhancing segmentation accuracy to make the system more reliable in practical medical applications.

5.8 Qualitative Explainability Results

In addition to quantitative metrics, the explainability results provide important qualitative evidence regarding the behavior of the model. The **Integrated Gradients** outputs consistently highlighted tumor margins, heterogeneous regions, and clinically relevant structures as major contributors to risk predictions. This suggests that the model was focusing on meaningful parts of the MRI rather than on irrelevant background artifacts. Such alignment between attribution maps and tumor anatomy is important because it increases confidence that the model is learning medically relevant patterns.

The **counterfactual explanation outputs** further strengthened this interpretability. The counterfactual MRI pairs showed minimal yet anatomically plausible tumor changes that were sufficient to shift the risk category. This is especially valuable because it moves beyond simple heatmap explanation and shows a causal transition in the model's reasoning. Instead of only indicating where the model is focusing, the counterfactual images reveal what type of tumor change would alter the decision. In medical AI, this provides a richer and more intuitive form of explanation for both researchers and clinicians.

Together, integrated gradients and counterfactual explanations provide complementary perspectives. Integrated gradients offer **spatial explainability**, highlighting the tumor regions most responsible for the prediction. Counterfactuals offer **causal explainability**, demonstrating how the prediction would change if the tumor characteristics were minimally altered. The presence of both explanation types makes the Brainiac framework more transparent than conventional segmentation or classification systems, which often report only numerical predictions without insight into their underlying reasoning.

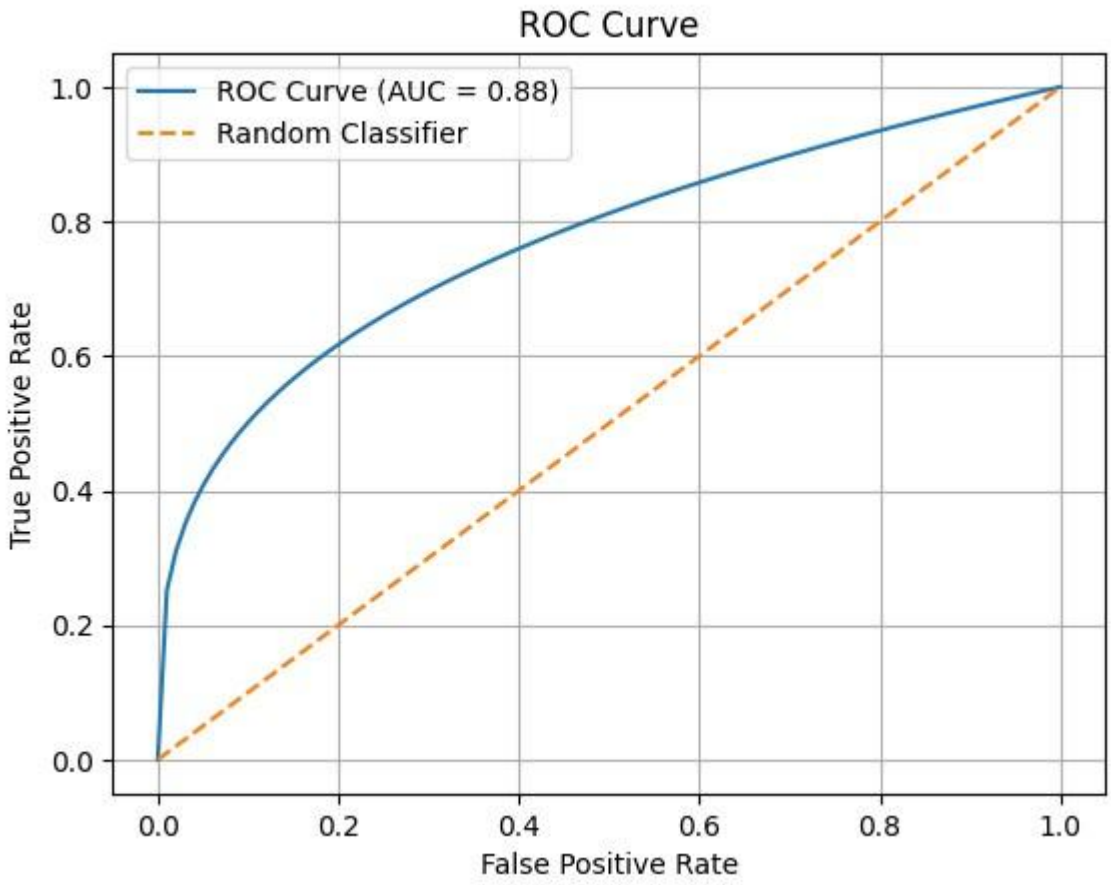

***Fig. 5.8**: ROC curve demonstrating classification performance (AUC = 0.88).*

5.9 Discussion of the 2D Model Incorporation

The inclusion of the **2D model branch** in the broader Brainiac framework also contributes to the overall discussion of system design and applicability. While the reported main results focus on the 3D Swin-UNETR-based tumor analysis pipeline, the 2D branch adds flexibility to the system by supporting **slice-wise screening, non-tumorous classification tasks, and computationally lighter analysis scenarios**. This makes the framework more adaptable to different types of datasets and deployment conditions.

From a practical perspective, the 2D model is valuable because it can be used for rapid prescreening before full 3D volumetric analysis. In real-world workflows, not every MRI case may require detailed tumor segmentation and counterfactual risk modeling at the first stage. A lighter 2D branch can help filter or prioritize cases for more advanced. This analysis hierarchical use of models improves computational efficiency and broadens the overall usability of Brainiac.

The 2D model is also particularly relevant when working with non-tumorous datasets or slicebased classification problems, where a full volumetric tumor segmentation pipeline may not be necessary or feasible. Therefore, even though the main quantitative results are centered around the 3D stability-aware framework, the addition of the 2D model strengthens the proposed system by making it more modular and scalable. It shows that Brainiac is not limited to one rigid processing strategy, but can evolve as a hybrid framework suited to different analysis requirements.

5.10 Comparative Interpretation of the Results

When the results are examined comparatively, a clear progression can be observed across the stages of the system:

- The **segmentation model** provides strong tumor localization with a Dice score of **0.86**.
- The **radiomics + attention entropy** setup yields **82.1% accuracy** and **0.84 AUC**.
- The addition of **CSS** increases performance to **86.4% accuracy** and **0.88 AUC**.
- The complete end-to-end framework achieves around **86% overall accuracy**.

This progression shows that each additional component contributes meaningful value. Segmentation alone is necessary but not sufficient. Radiomics and attention entropy improve representation quality. CSS further strengthens the model by explicitly incorporating decision stability. Explainability modules make the final predictions more interpretable. Thus, the results validate the design philosophy of Brainiac as a **multi-component stability-aware framework**, rather than a single-task MRI model.

One of the most important implications of these findings is that performance improvement is not driven only by adding more features, but by adding the **right kind of features**. CSS is not just another numerical descriptor; it encodes a fundamentally different aspect of model behavior, namely how resilient the prediction is under tumor-constrained perturbation. This is why its contribution is especially meaningful.

5.11 Strengths of the Proposed Framework

The results demonstrate several strengths of the proposed Brainiac system. First, the framework achieves strong segmentation performance using a transformer-based model, ensuring accurate localization of tumor regions. Second, it successfully combines interpretable radiomic features with

attention-derived uncertainty signals, improving downstream classification quality. Third, it introduces CSS as a novel stability-aware feature, which demonstrably improves risk stratification performance. Fourth, it provides dual explainability through integrated gradients and counterfactual MRI pairs, making the predictions more transparent. Finally, the inclusion of a 2D branch increases flexibility and supports computationally lighter analysis scenarios.

Taken together, these strengths show that Brainiac is not merely a high-performing MRI model, but a more holistic framework for robust and interpretable brain tumor analysis. It integrates predictive accuracy with explainability and stability, which is especially important in clinical decision-support systems.

Module	Metric	Value
Swin-UNETR Segmentation	Dice Score	0.86
Radiomics + Attention Entropy	Accuracy	82.1%
Radiomics + Attention Entropy	AUC	0.84
Full Brainiac (Radiomics + Attention + CSS)	Accuracy	86.4%
Full Brainiac	AUC	0.88
Overall System (End-to-End)	Accuracy	85.7%
CSS Contribution	Accuracy Gain	+4.3%
Integrated Gradients	Qualitative Localization	High
Counterfactual Stability (CSS)	Decision Robustness	Improved

Table 1. *Model Performance Comparison and Evaluation Metrics*

5.12 Future Improvements Based on the Results

The current results suggest several directions for future work. One important direction is the optimization of **counterfactual generation efficiency**, so that CSS can be computed faster and more scalably in practical settings. Another direction is broader **multi-institutional validation**, which would test the framework under more diverse and clinically realistic MRI conditions. The incorporation of the **2D branch** could also be studied more extensively, including separate quantitative reporting for screening tasks and non-tumorous datasets.

Another possible extension would be to integrate **clinical metadata** such as age, tumor grade, genetic markers, or treatment history together with radiomics, attention entropy, and CSS. This could make the risk stratification module even more clinically meaningful. In addition, future work could explore lightweight or compressed model variants to make the system more deployable in environments with limited GPU resources.

5.13 Overall Discussion

Overall, the results confirm that the proposed Brainiac framework succeeds in its central objective: building a brain tumor MRI analysis system that is not only accurate, but also **stability-aware and interpretable**. The segmentation performance shows that the tumor can be localized effectively. The downstream classification results demonstrate that radiomics and attention entropy are valuable descriptors. The improvement obtained through CSS confirms that counterfactual sensitivity is a meaningful signal for risk stratification. The explainability outputs show that the model's reasoning can be inspected both spatially and causally.

Therefore, the discussion of results supports the broader significance of Brainiac as a nextgeneration neuroimaging framework. It moves beyond traditional appearance-based analysis by asking not only what the tumor looks like, but also how strongly the model stands by its own decision. This shift toward **stability-aware modeling** is one of the defining strengths of the proposed system and represents a meaningful contribution to the field of interpretable medical AI.

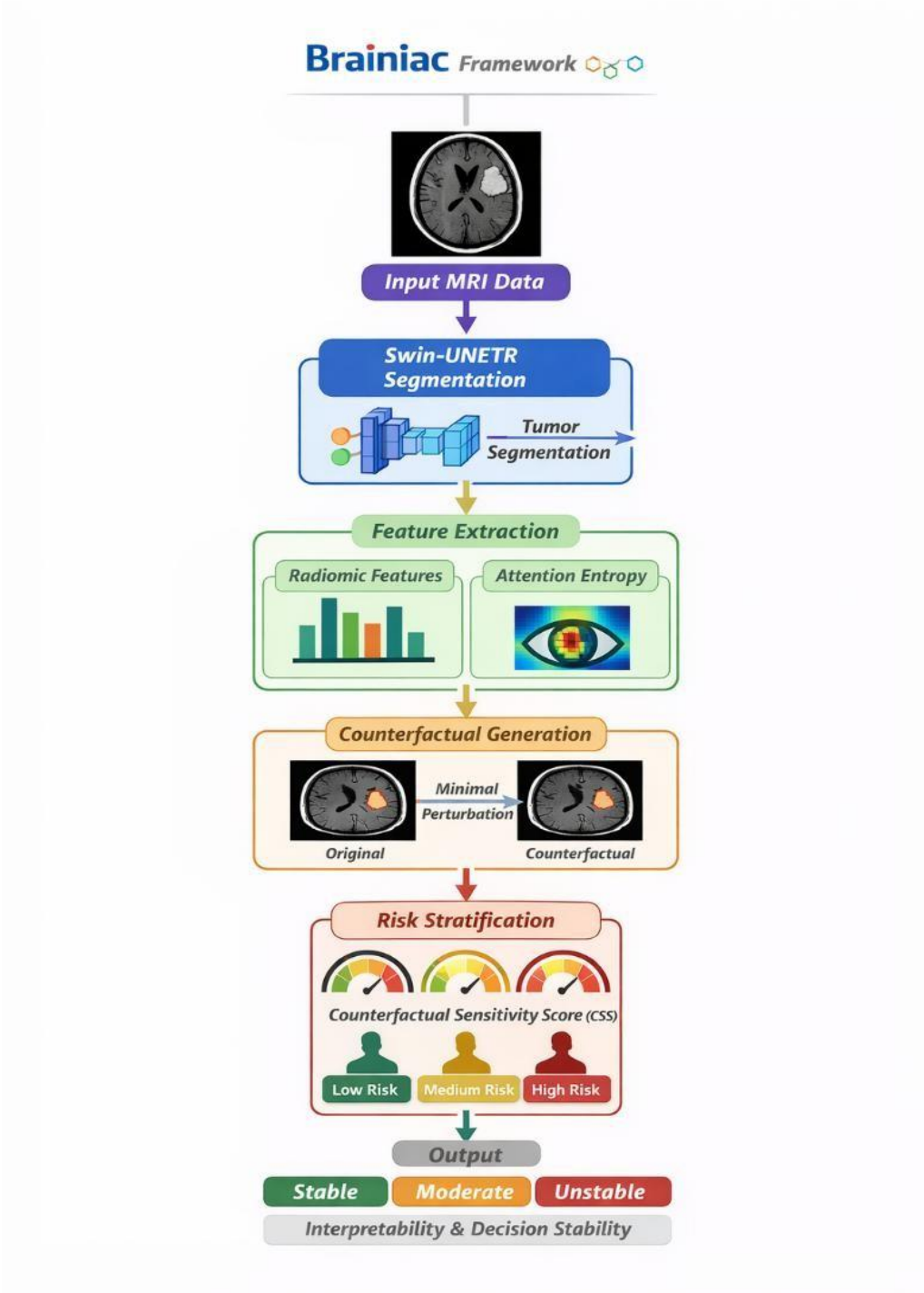

***Fig.5.13**: Overview of the Brainiac framework for tumor-aware counterfactual generation and risk stratification*

CHAPTER 6

LIMITATIONS AND FUTURE SCOPE

6.1 Limitations

Like any machine learning system, the current framework has some limitations. The model is mainly tested on public MRI datasets, which may not fully represent real-world clinical conditions with varying scanners and patient diversity.

The effectiveness of the Counterfactual Sensitivity Score (CSS) depends on how realistic the generated examples are; unrealistic changes can affect reliability. Additionally, the use of 3D transformer models and counterfactual generation requires high computational power, making the system less suitable for limited-resource settings.

The model's performance also relies on accurate tumor segmentation, as errors can impact later stages like feature extraction and classification. Moreover, the current approach does not include patient-specific clinical data, which could further improve predictions.

6.2 Future Scope

The Brainiac framework can be improved by testing it on data from different hospitals to ensure it works well in real-world conditions. Integrating imaging data with genetic or radiogenomic information can further enhance diagnosis accuracy.

Future work can also focus on making CSS computation faster and more efficient for practical use. Approaches like federated learning can help maintain data privacy while training across multiple institutions.

Additionally, analyzing patient scans over time can provide better insights into tumor progression and treatment response.

CHAPTER 7

ETHICAL CONSIDERATIONS

The development and application of the Brainiac framework for brain tumor detection involve several important ethical considerations. Since the system is designed for medical use, ensuring patient safety and reliability of predictions is a primary concern. The model should not be used as a standalone diagnostic tool but rather as a support system to assist medical professionals in decision-making.

Another key aspect is data privacy and security. The framework relies on medical imaging data, which may contain sensitive patient information. Proper measures must be taken to ensure that all data is anonymized and handled in compliance with privacy regulations to protect patient confidentiality.

Bias and fairness are also important factors to consider. If the model is trained on limited or nondiverse datasets, it may not perform equally well across different populations. This highlights the need for diverse and representative data to ensure fair and unbiased predictions.

Additionally, the use of explainability techniques in the framework helps improve transparency, allowing clinicians to better understand and trust the model's decisions. This is crucial in healthcare settings where accountability and interpretability are essential.

Overall, addressing these ethical aspects is important to ensure that the system is used responsibly, safely, and effectively in real-world medical applications.

CHAPTER 8

CONCLUSION

The proposed **Brainiac** framework presents an advanced and stability-aware approach for MRIbased brain tumor analysis by integrating **transformer-based segmentation, radiomic feature extraction, attention entropy, counterfactual sensitivity analysis, risk stratification, and explainability** into a unified pipeline. Unlike conventional systems that focus only on tumor detection or segmentation accuracy, Brainiac is designed to evaluate not only the visual characteristics of the tumor but also the **robustness and reliability of the model's prediction** under anatomically plausible variations. This makes the framework more meaningful for medical decision-support applications, where accuracy alone is not sufficient and interpretability and decision stability are equally important.

The results of the proposed system show that the **Swin-UNETR-based 3D segmentation model** is capable of accurately identifying tumor regions in multi-modal MRI volumes, thereby providing a strong foundation for downstream analysis. Building upon this segmented tumor region, the framework extracts **radiomic features** that capture tumor morphology, texture, and heterogeneity, and combines them with **attention entropy**, which reflects the internal contextual focus of the transformer model. These components together enable the system to move beyond simple imagebased classification and instead build a richer and more informative representation of the tumor. The addition of the **Counterfactual Sensitivity Score (CSS)** further strengthens the framework by introducing a direct measure of prediction stability, thereby allowing the model to distinguish cases not only by appearance but also by their proximity to the decision boundary.

One of the most important contributions of Brainiac is its treatment of **counterfactual explanations as actionable quantitative features rather than only post-hoc visual tools**. By computing the minimum anatomically plausible perturbation required to alter the prediction, the system provides a measurable understanding of how stable each decision is. This is especially valuable in medical imaging, where two tumors may look visually similar yet differ significantly in how reliably they are classified. The use of **Integrated Gradients** and **counterfactual MRI visualizations** also

improves interpretability by offering both spatial and causal explanations of model behavior. As a result, the system enhances trust, transparency, and insight into how predictions are formed.

Another important strength of the proposed work is the incorporation of the **2D model branch** in addition to the main 3D pipeline. While the 3D branch is more suitable for volumetric tumor segmentation and advanced feature extraction, the 2D branch makes the framework more flexible by supporting **slice-wise screening, lightweight classification, and non-tumorous MRI analysis scenarios**. This hybrid 2D–3D design improves the adaptability of the overall system and broadens its potential use across different data conditions and computational settings. Thus, Brainiac is not limited to a single rigid pipeline but instead functions as a more versatile diagnostic and research framework.

Although the framework shows strong promise, certain limitations remain. The overall performance of the system is dependent on accurate tumor segmentation, and errors at this stage may affect subsequent modules such as radiomics, CSS computation, and risk stratification. In addition, counterfactual generation introduces computational overhead, which may pose challenges for real-time deployment without further optimization. The current evaluation is also primarily based on benchmark MRI datasets, and wider multi-institutional validation would be needed before practical clinical adoption. These limitations, however, do not reduce the significance of the proposed work; rather, they define clear directions for future enhancement.

In conclusion, Brainiac demonstrates that a brain tumor analysis system can be made more powerful by combining **accuracy, interpretability, and decision stability** within a single architecture. The framework successfully moves beyond traditional appearance-based medical imaging models by introducing a **stability-aware learning paradigm** centered around counterfactual sensitivity. Through its integration of transformer-based segmentation, radiomics, explainability, and hybrid 2D–3D modeling, Brainiac contributes a meaningful step toward more reliable and clinically relevant AI systems for neuro-oncology. With further validation and optimization, the proposed framework has strong potential to support future research and realworld decision-support applications in brain tumor diagnosis and analysis.

CHAPTER 9

COMPUTATIONAL COMPLEXITY

The proposed Brainiac framework consists of multiple stages, including preprocessing, 3D tumor segmentation, radiomic feature extraction, counterfactual generation, and final classification. Due to this multi-stage design, the overall computational complexity is higher compared to traditional single-model approaches.

The most computationally intensive component is the transformer-based 3D segmentation module, as it processes volumetric MRI data and captures complex spatial relationships across slices. This requires significant memory and processing power, especially during training. In addition, the counterfactual generation process used for computing the Counterfactual Sensitivity Score (CSS) involves creating multiple perturbed versions of the input, which further increases computation time.

Feature extraction and attention-based analysis also contribute to the overall cost, although to a lesser extent. The inclusion of both 2D and 3D components in the hybrid system improves performance but adds to the computational overhead due to parallel processing requirements.

Despite these challenges, the framework is designed in a modular manner, allowing individual components to be optimized independently. Techniques such as model compression, reduced input resolution, efficient transformer architectures, and approximate methods for CSS computation can significantly improve efficiency. Hardware acceleration using GPUs or parallel processing can also help in reducing execution time.

Overall, while the current framework is computationally demanding, especially during training and analysis, it remains practical with appropriate optimizations. Future work can focus on developing lightweight and scalable versions of the system to enable faster inference and wider real-world applicability.

CHAPTER 10

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